



Alveolar bone-1 and 2



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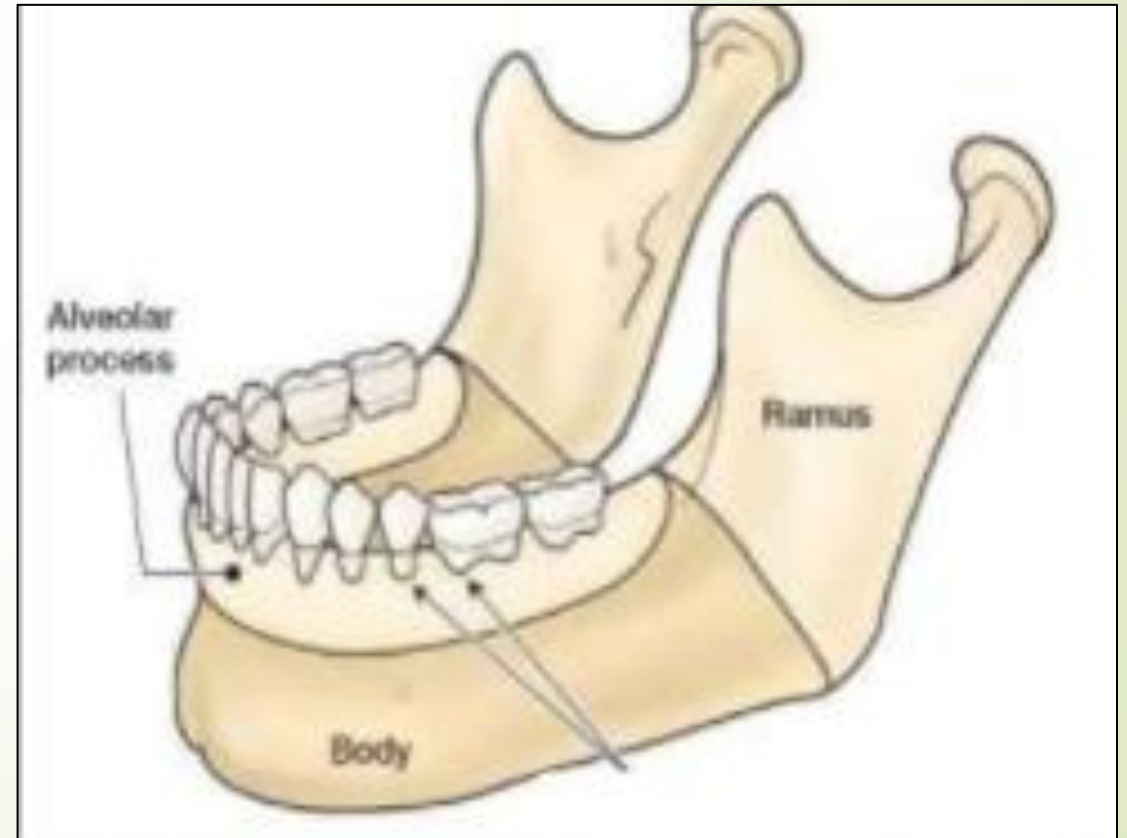
BDS, MDS, Oral and Maxillofacial Pathology

Learning outcomes

- **Discuss the physical properties and chemical composition of alveolar bone**
- **Describe classification and main structural features of alveolar bone**
- **Histology of alveolar bone and explain the functions of osteoblast, osteocytes and osteoclasts**
- **List the functions of alveolar bone**

INTRODUCTION

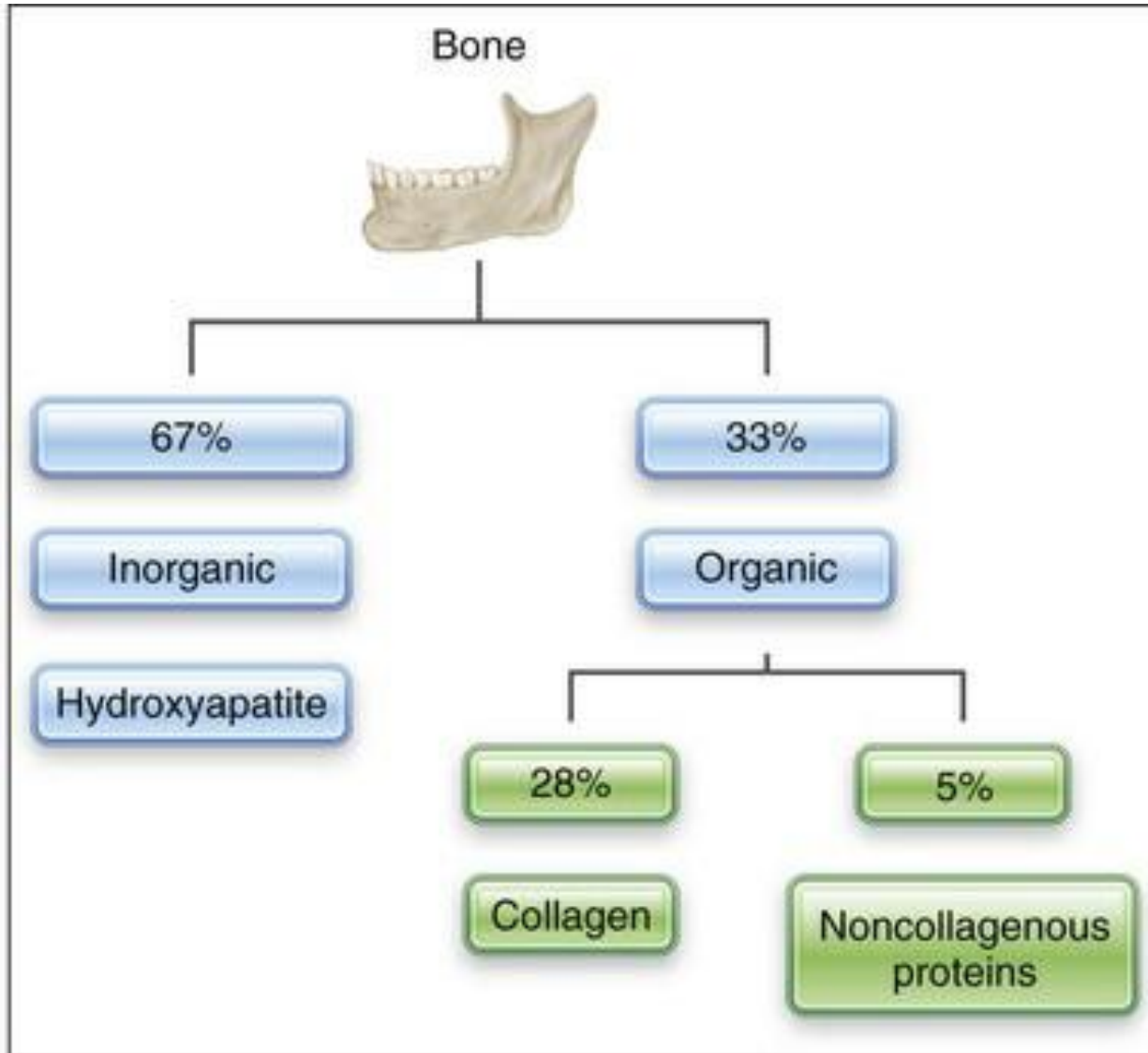
- ➔ The alveolar process is defined as that part of the maxilla and the mandible that forms and supports the sockets of the teeth
- ➔ If the teeth are lost the alveolar bone disappears



Physical properties

- **Bone is a calcified connective tissue**
- **It is highly vascular**
- **High remodeling capacity**

Chemical Composition of bone



Collagen

Type I > 95%

Type V < 5%

Non-collagenous proteins

Osteocalcin

Osteopontin

Osteonectin

Bone sialoprotein

Classification of bone

→ Bone may be classified in several ways:

Developmentally	Histologically	By shape
Endochondral bone	Compact (Cortical) Bone	Long bones
Intramembranous bone	Spongy (Cancellous) Bone	Short bones
		Flat bones
		Irregular bones

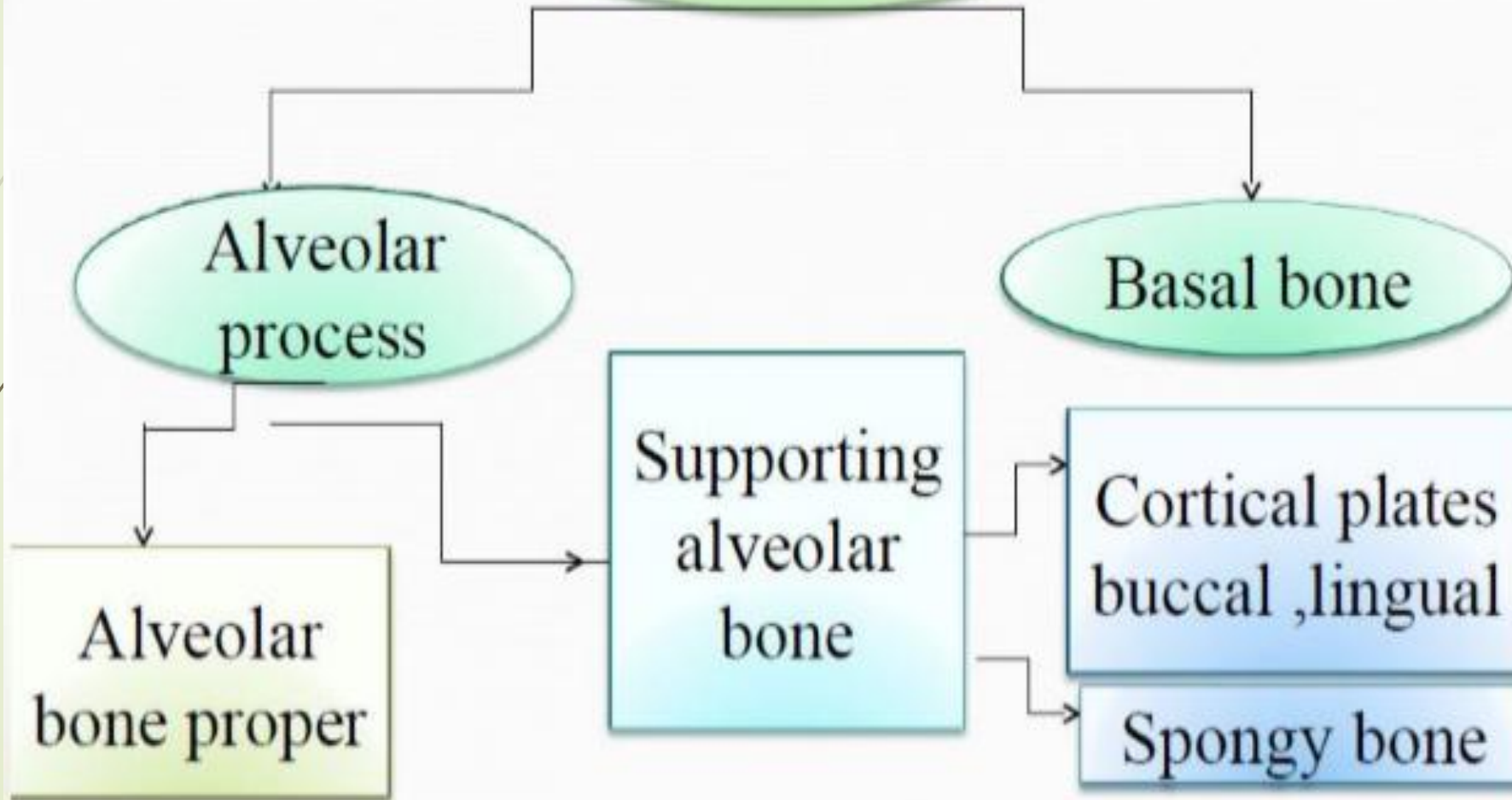


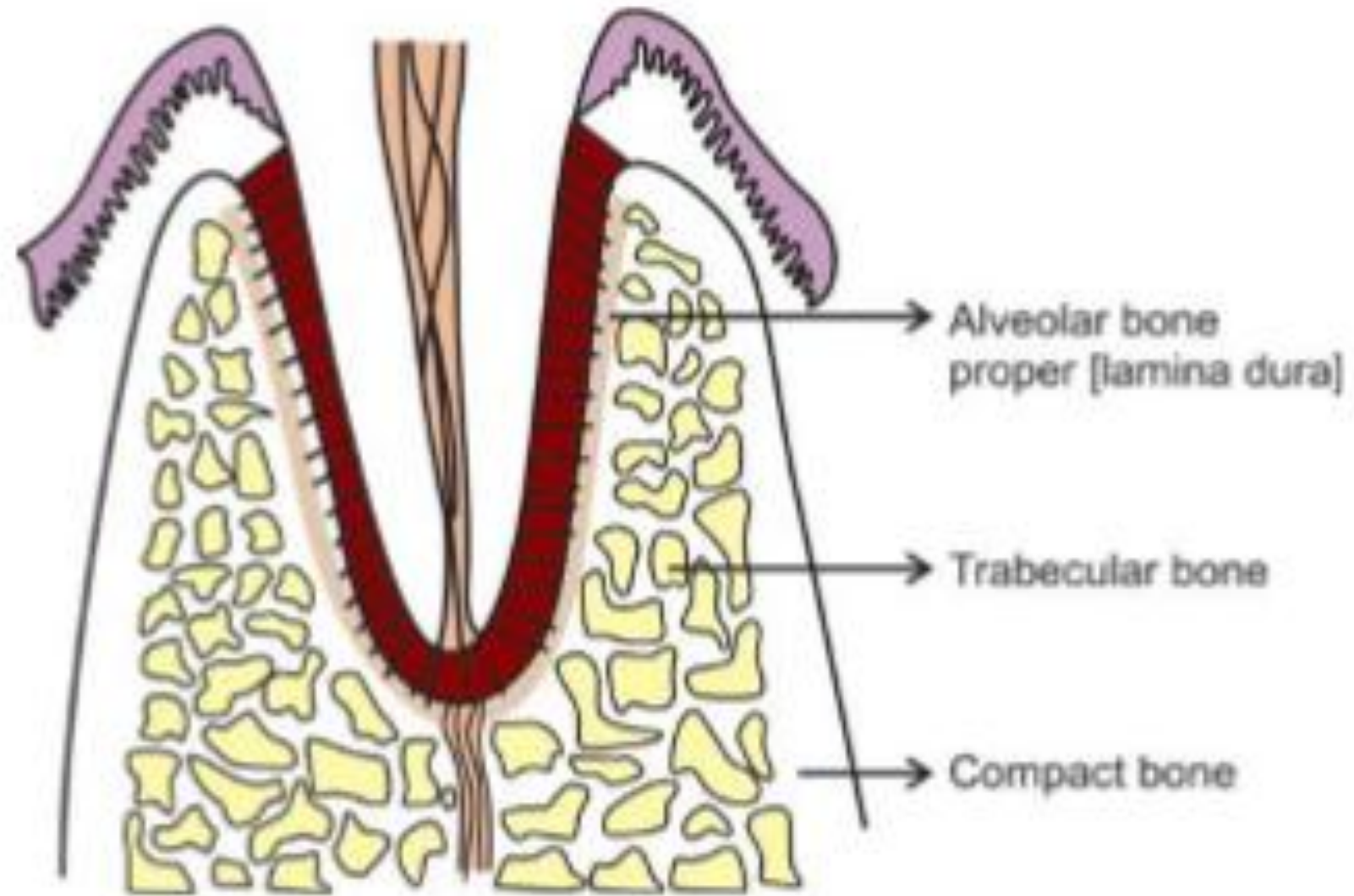
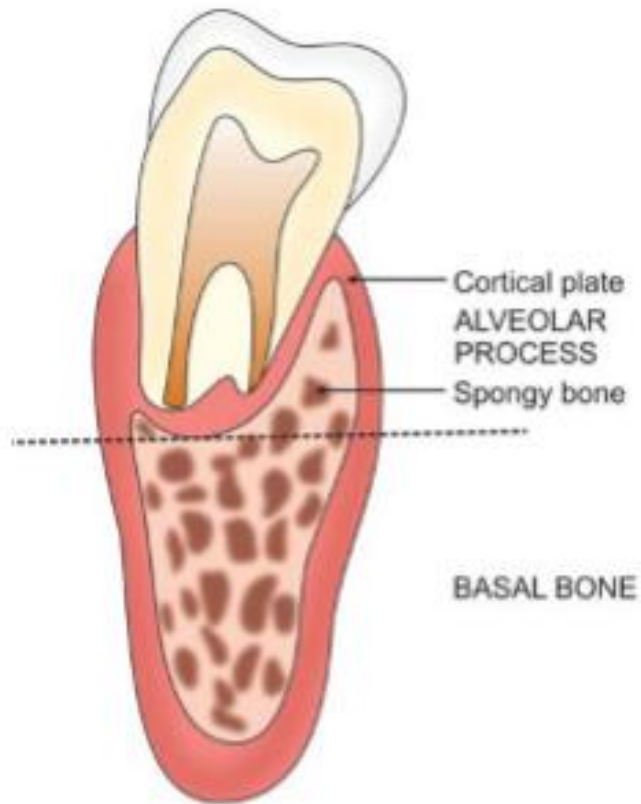
Structural features of alveolar bone

Structure of alveolar bone

- The alveolar process consists:
 - Bone lining the alveolus (alveolar bone proper)**
 - Supporting alveolar bone consisting of**
 - (i) outer (buccal and lingual) cortical plates
 - (ii) a central spongiosa (cancellous bone/spongy bone)

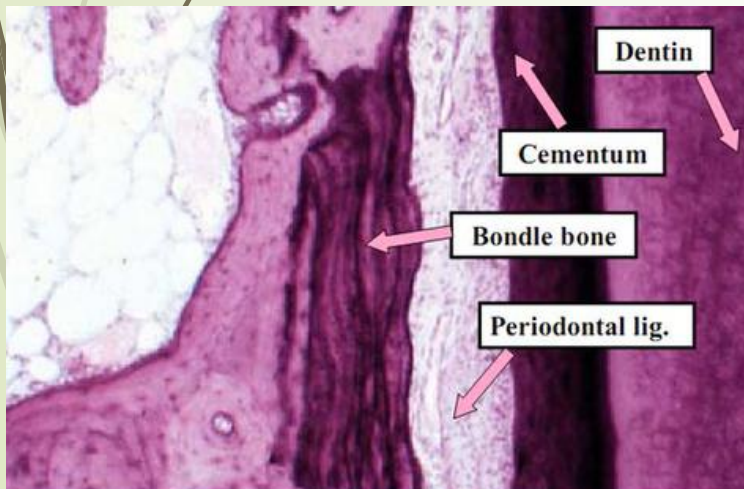
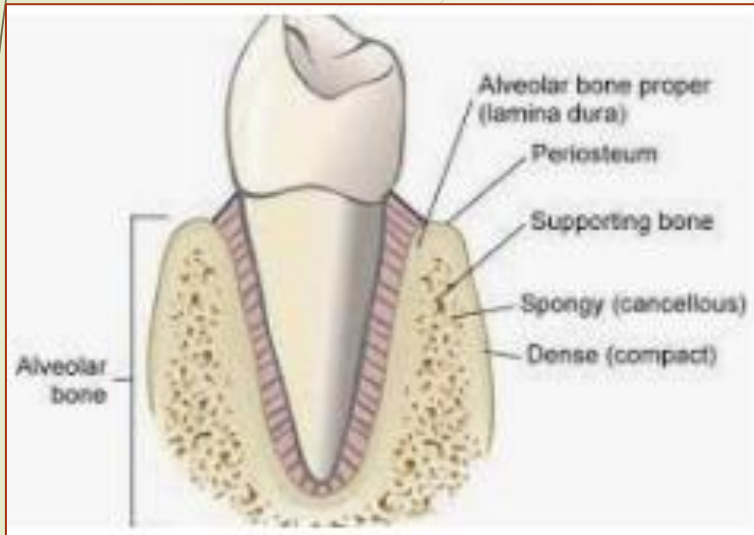
Jaw bones





Alveolar bone structure

Alveolar bone proper (ABP)

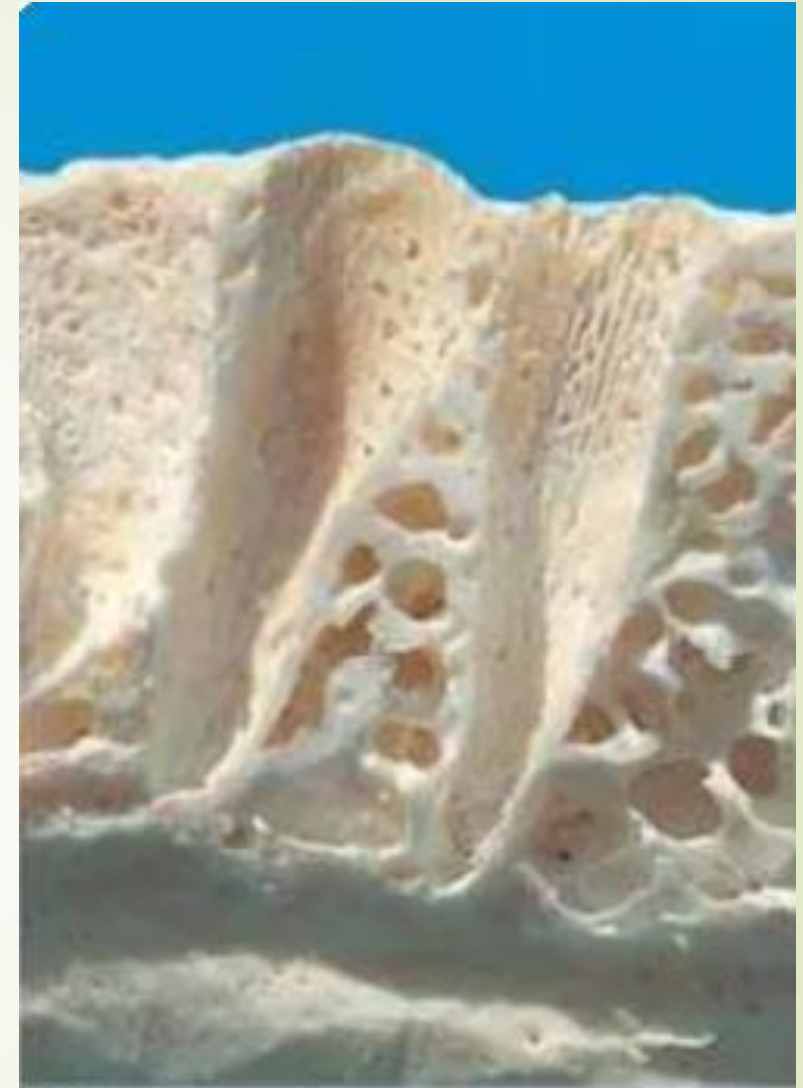
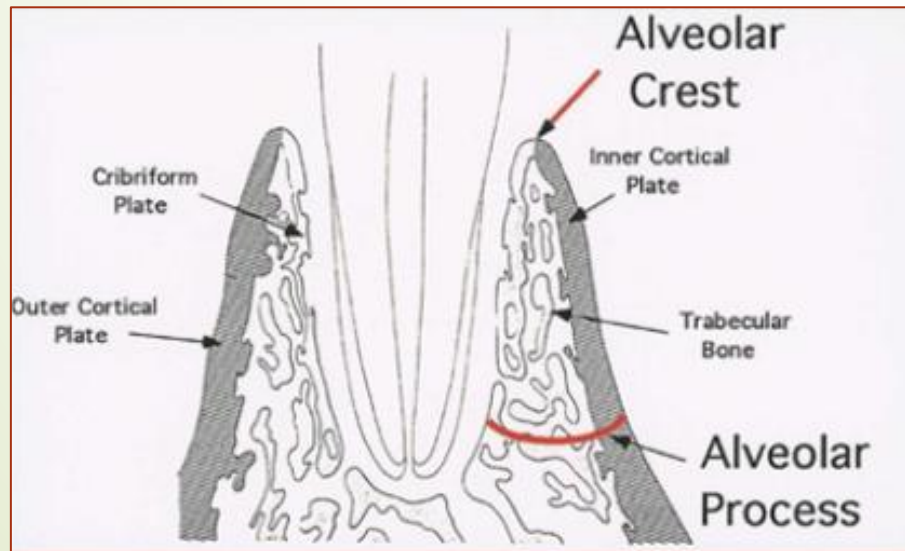


- ABP is made up of:
 1. **Bundle bone**--bone directly lining the socket into which the principal fibers of the periodontal ligament are anchored
 2. **Lamellated bone**—bone immediately adjacent to the bundle bone and contains osteons
- It is about **0.1–0.4 mm** thick
- Radiographically--**lamina dura**; increased radiopacity, due to the presence of thick bone without trabeculations

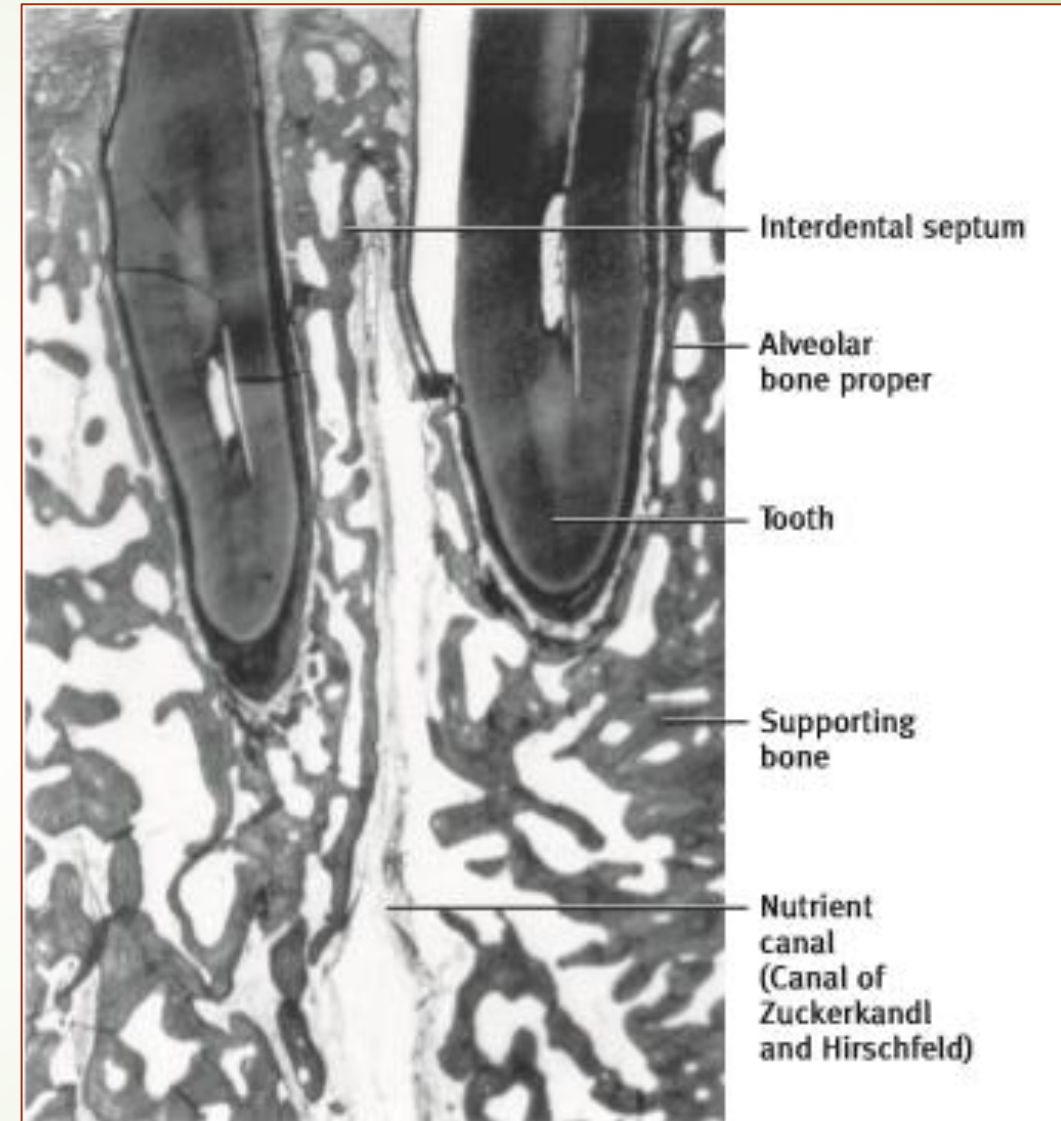


➤ The alveolar bone proper, which forms the inner wall of the socket is perforated by many openings that carry branches of the interalveolar nerves and blood vessels into the periodontal ligament--called the **cribriform plate**

➤ Bone between the teeth is called **interdental septum** and is composed entirely of cribriform plate.

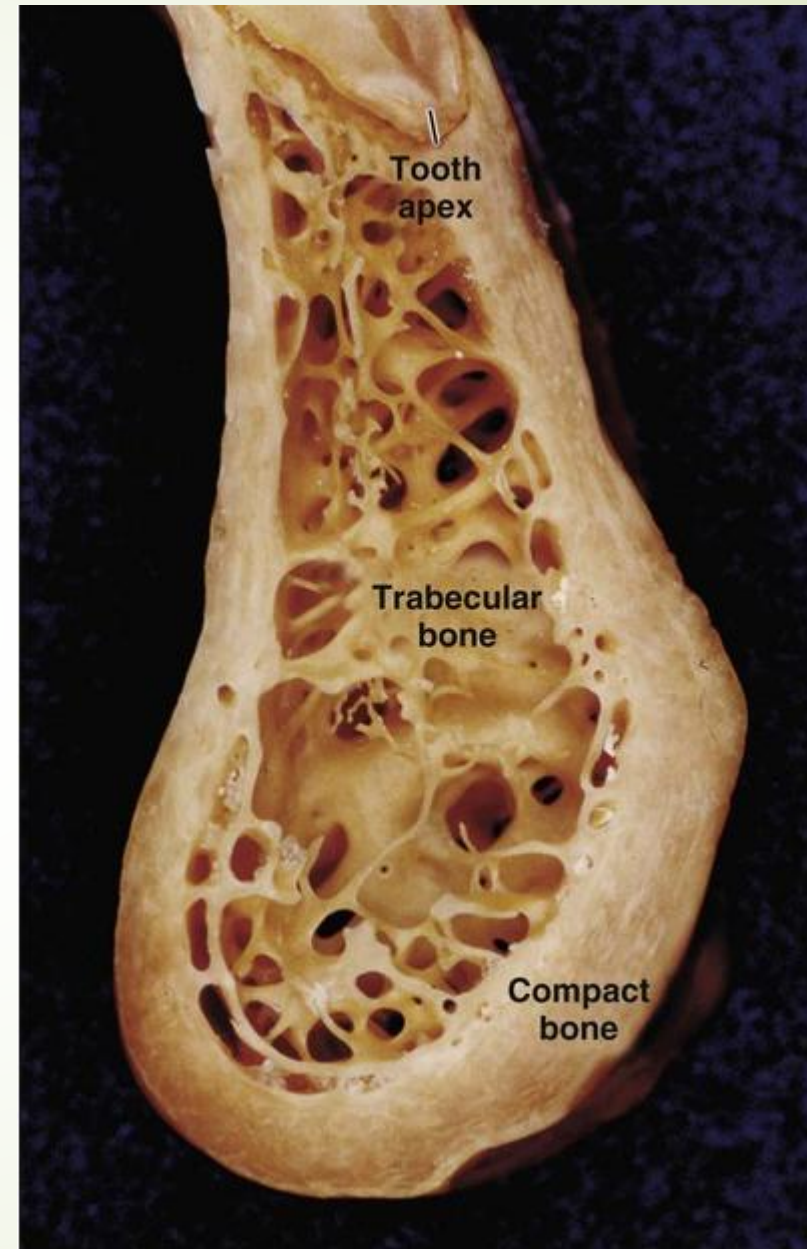


- The interdental and interradicular septa contain the perforating canals of *Zuckerkindl and Hirschfeld* (*nutrient canals*) which house the interdental and interradicular arteries, veins, lymph vessels and nerves



Supporting alveolar bone

- Consists of two parts:
- (a) Cortical plates
 - (b) Spongy bone

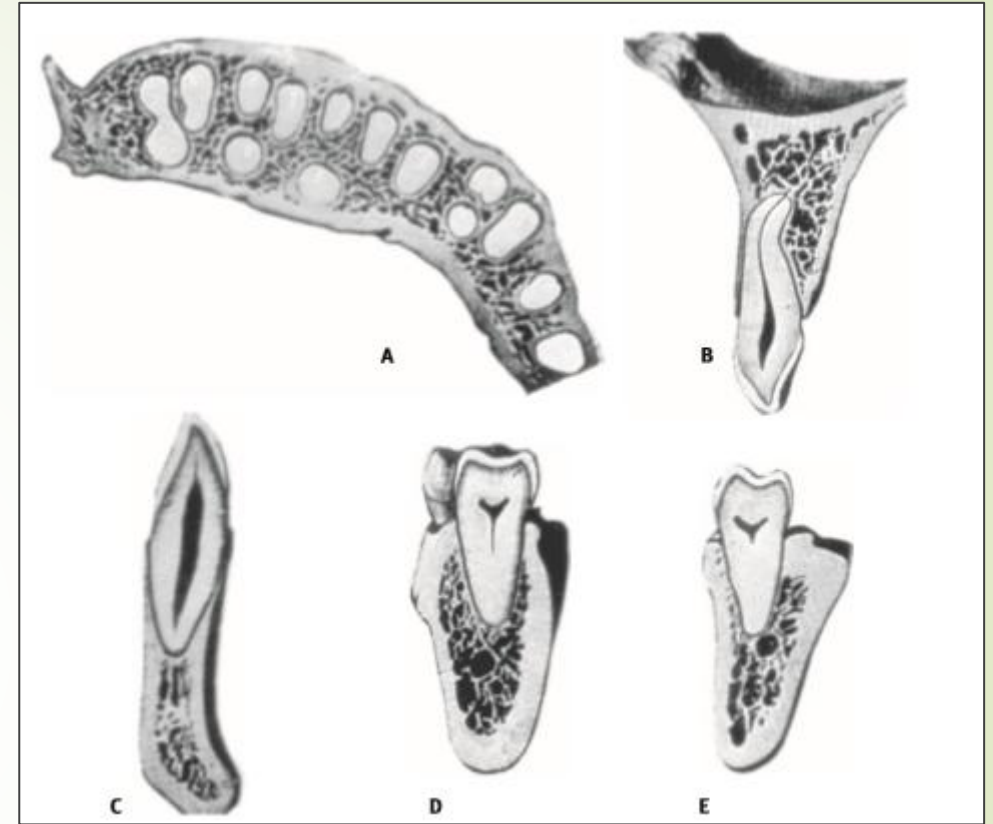


Cortical plates

- **Consist of compact bone and form the outer and inner plates of the alveolar processes**
- **Bone underlying the gingiva is the cortical plate**
- **Both cribriform plate and cortical plate are compact bone separated by spongy bone**
- **Histologically, the cortical plates consist of concentric lamellae and Haversian systems**

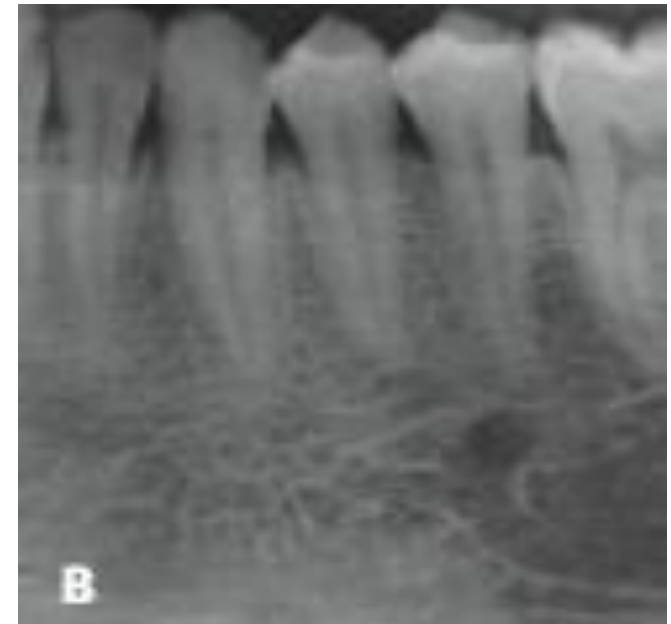
Spongy bone

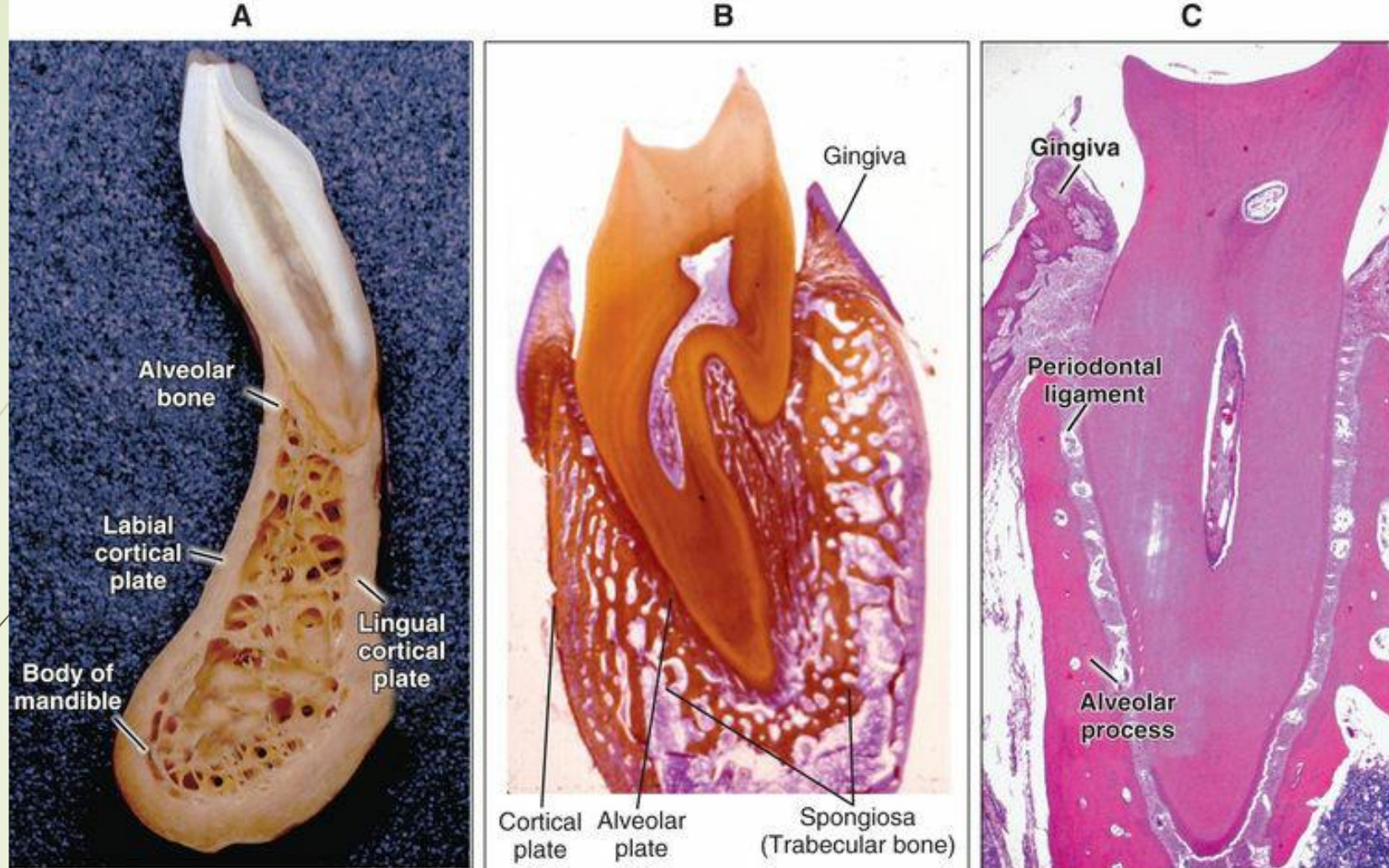
- **Spongy bone fills the area between the cortical plates and the alveolar bone proper. It contains trabeculae of lamellar bone**
- **The trabeculae contain osteocytes in the interior and osteoblasts or osteoclasts on the surface**



Gross relations of alveolar processes

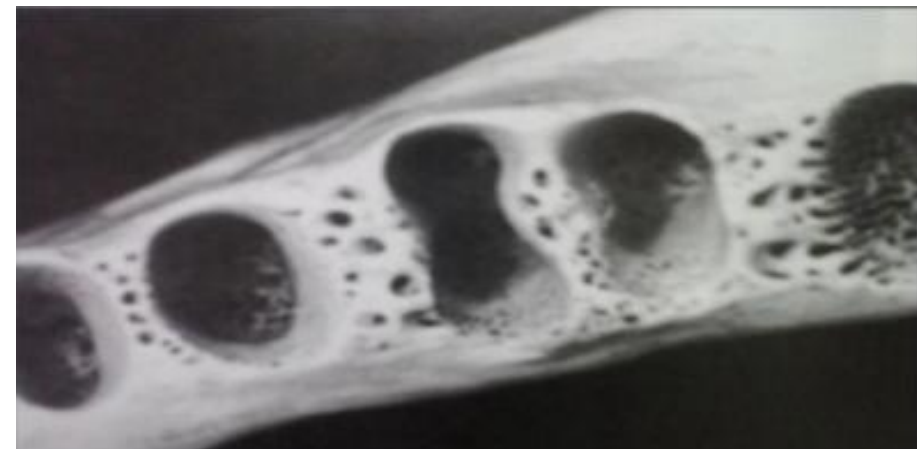
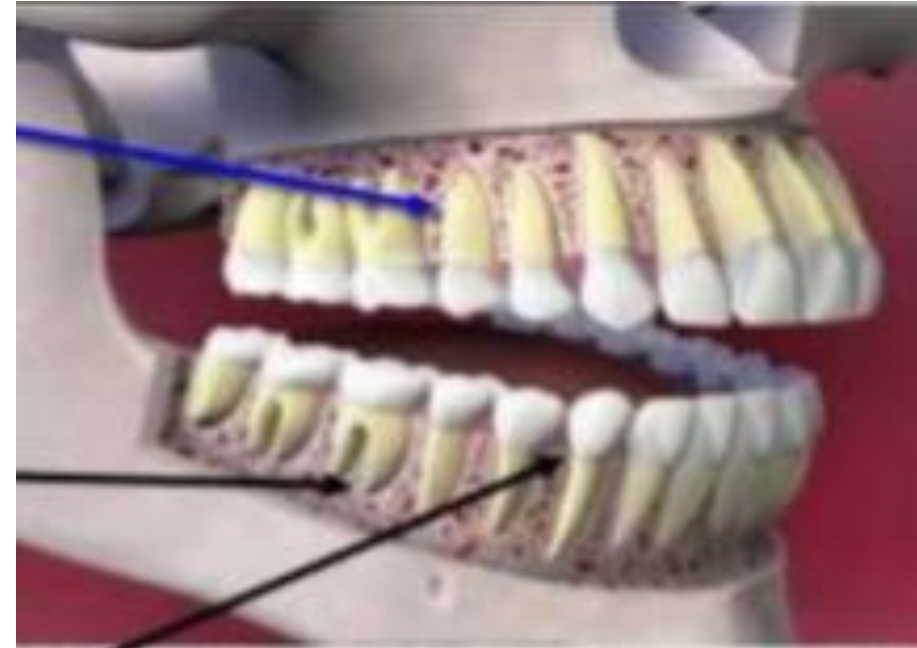
- ***Radiographs--classification of the spongiosa of the alveolar process into two main types***
- **Type I --- interdental and interradicular trabeculae are regular and horizontal in a ladder like arrangement**
- **Type II-- irregularly arranged, numerous, delicate interdental and inter-radicular trabeculae**



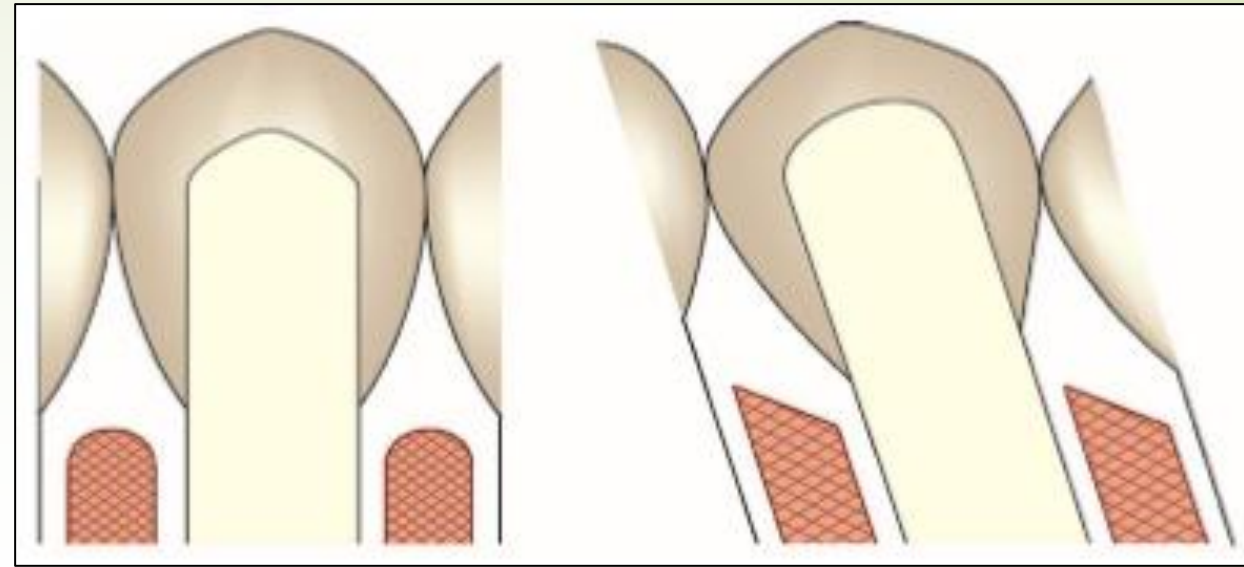


- (A)** Trabecular bone is found between the lingual cortical plate and alveolar bone in the region of the apical third of the root and in the body of the mandible
- (B)** a thick alveolar process with an abundant spongiosa (trabecular bone) between the cortical plates and alveolar bone
- (C)** a thin alveolar process lacking distinct trabecular bone.

- **Interdental septum**—Part of alveolar bone found between adjacent teeth
- **Inter-radicular bone**—Part of alveolar bone found between roots of teeth

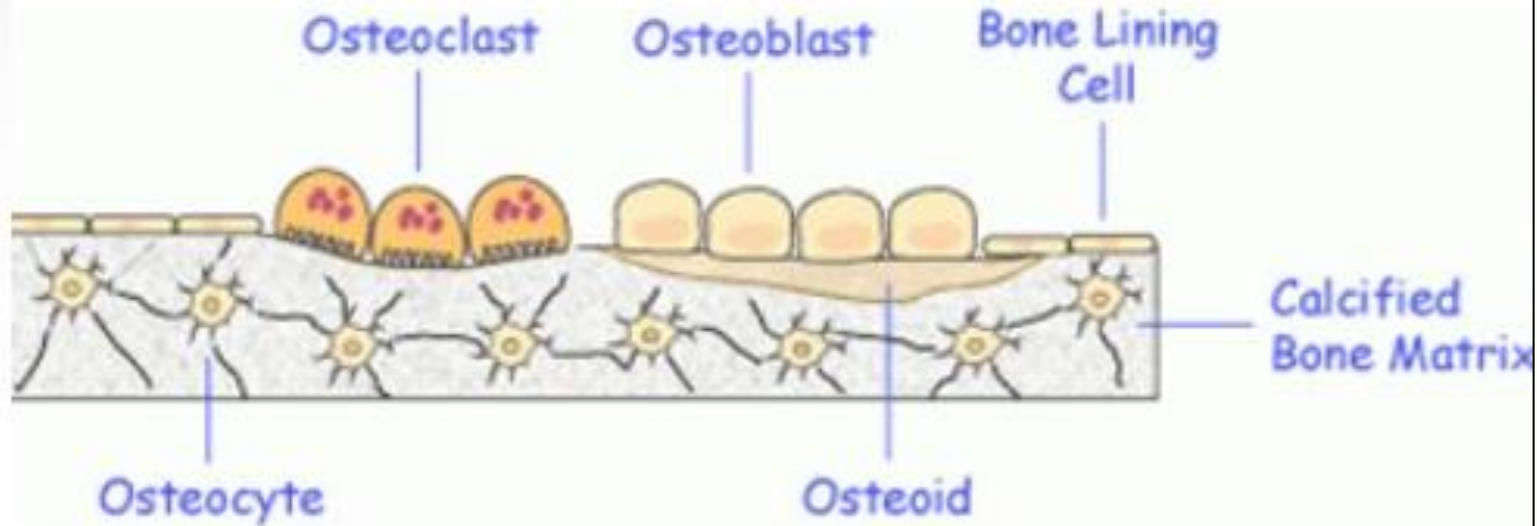


Crest of the alveolar septa



- The shape of the crest of the alveolar septa is dependent on the position of the adjacent teeth
- If the neighboring teeth are inclined, the alveolar crest is oblique
- Crest of the alveolar bone is usually 1.5 to 2 mm below the level of the cementoenamel junction on the tooth it surrounds

Bone cells

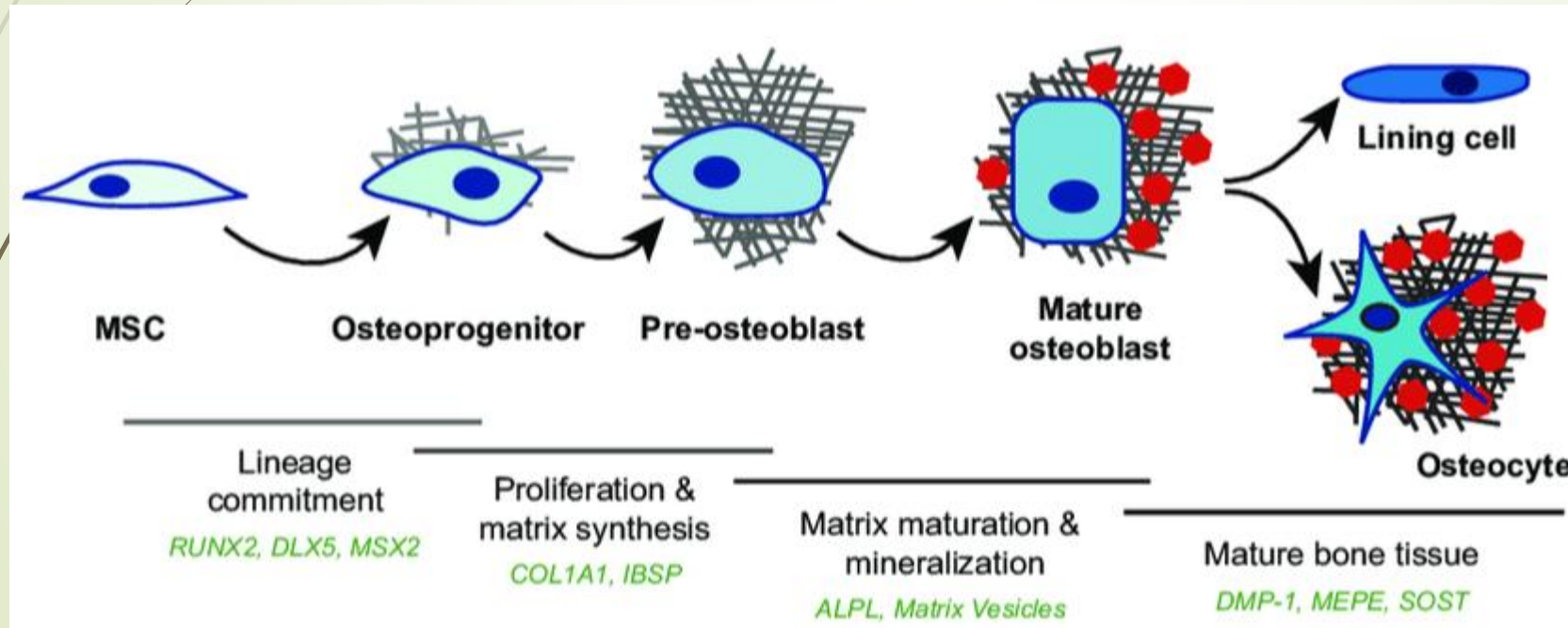


Bone cells

1. Osteoprogenitor cells
2. Osteoblasts
3. Osteoclasts
4. Osteocytes
5. Bone lining cells

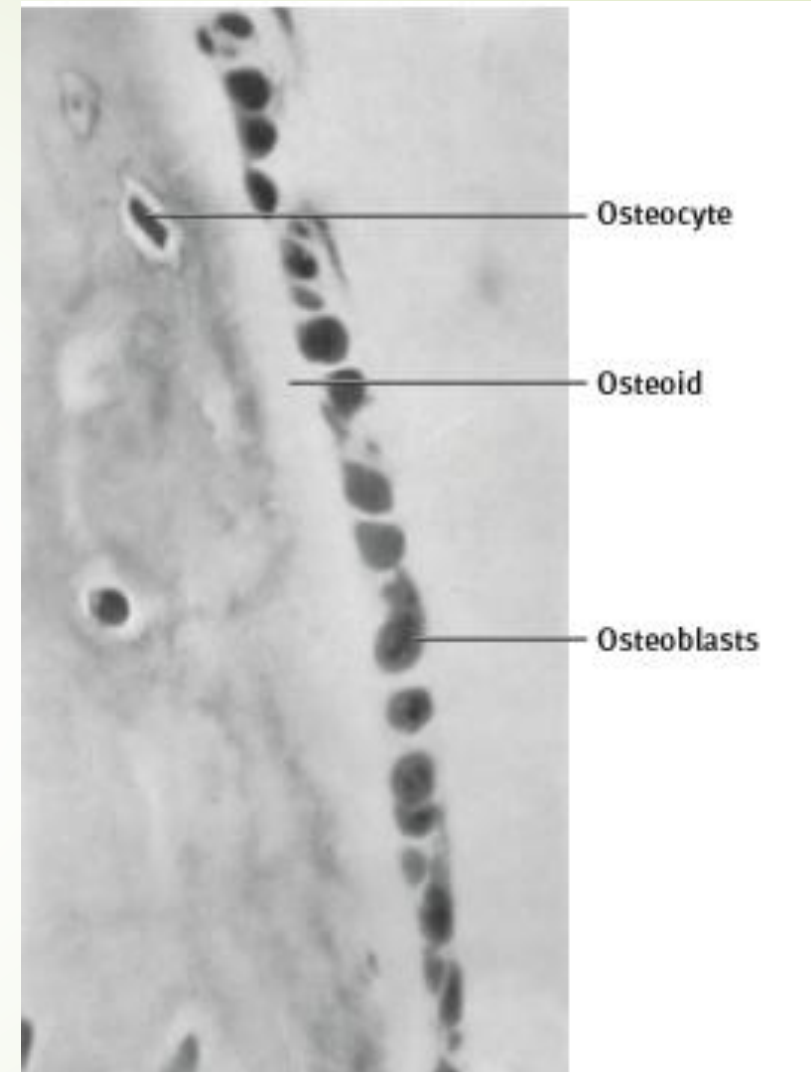
1. Osteoprogenitor cells--found in the membranous endosteum and periosteum.

- Functions--they differentiate into bone lining cells or osteoblasts when stimulated



2. Osteoblasts --mononucleated cells derived from osteoprogenitor cells of mesenchymal origin present in the bone marrow and other connective tissues

- Functions---** responsible for the synthesis and secretion of the macromolecular organic constituents of bone matrix



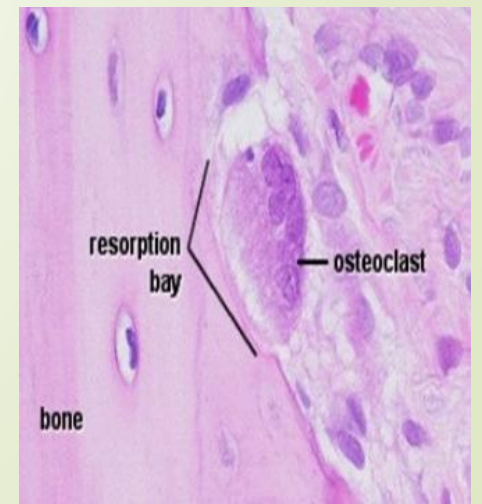
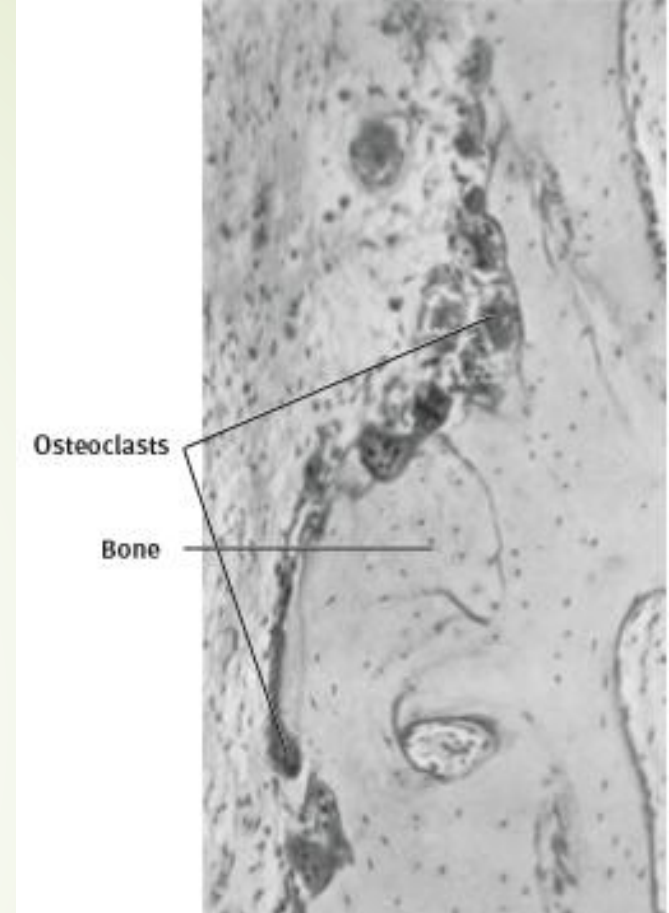
3. Osteoclasts -- large cell 40–100 μm in diameter with 15 to 20 closely packed nuclei

They lie in resorption bays called Howship's lacunae

Osteoclasts with many nuclei resorb more bone than osteoclasts with few nuclei

The presence of acid phosphatase distinguishes the osteoclast from other multinucleated giant cells

- Functions--removes bone tissue by removing the mineralized matrix of bone

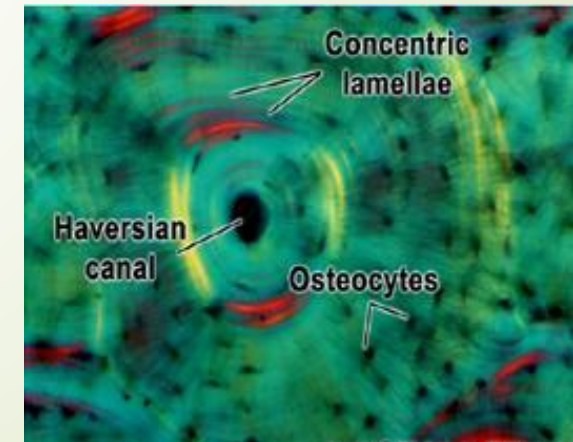
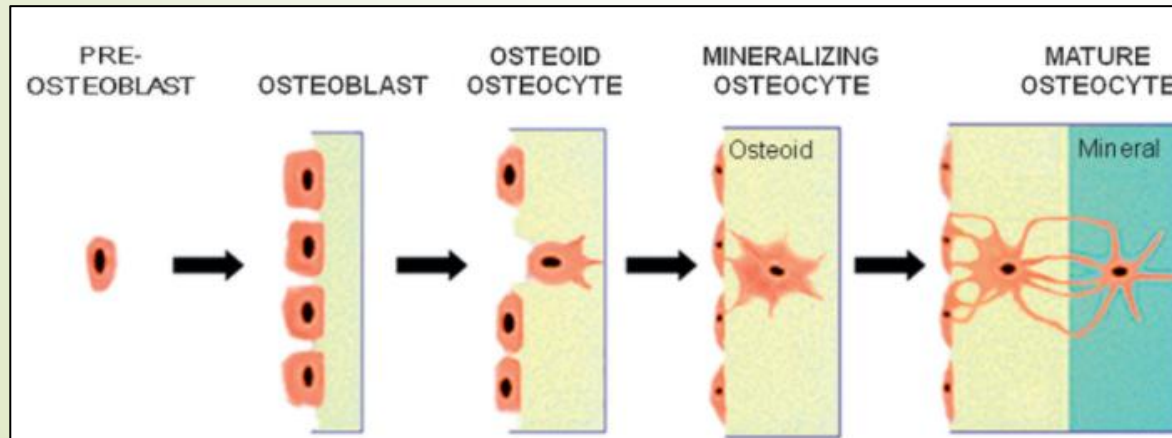


4. Osteocytes – as osteoblasts form the bone matrix, cells that get entrapped within the matrix

Osteocyte are present in a space around it, called the osteocytic lacuna

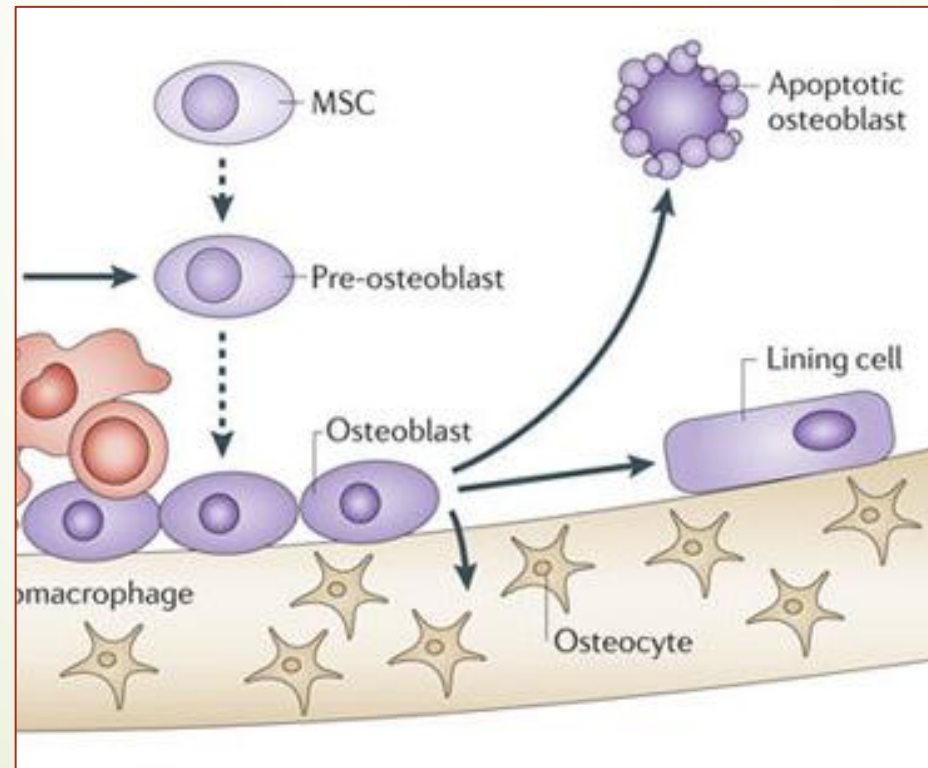
Narrow extensions of lacunae form channels --canaliculi which contain osteocytic processes

- **Function**--canaliculi penetrate the bone matrix and permit diffusion of nutrients, gases and waste products between osteocytes and blood vessels --maintains calcium and phosphate ion levels in blood



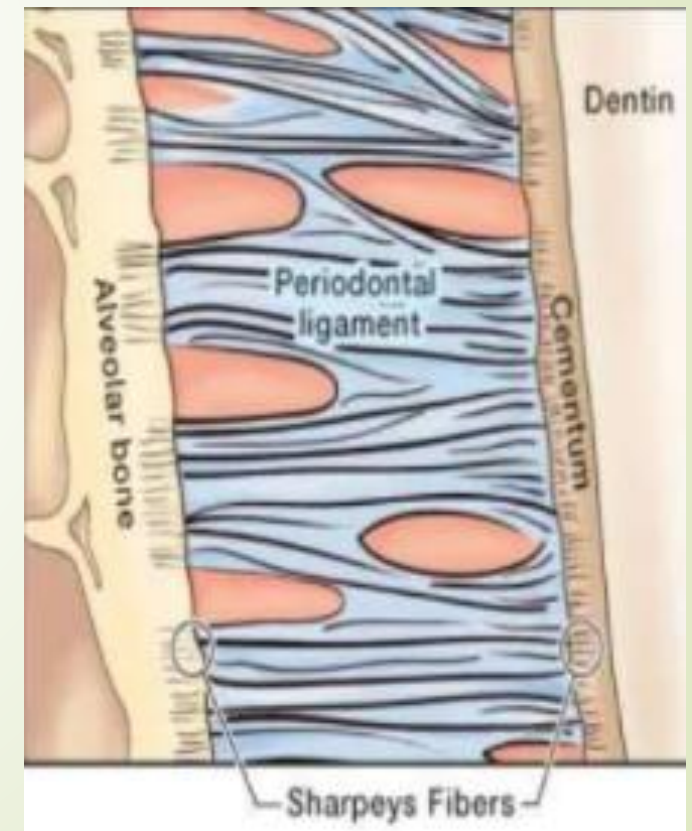
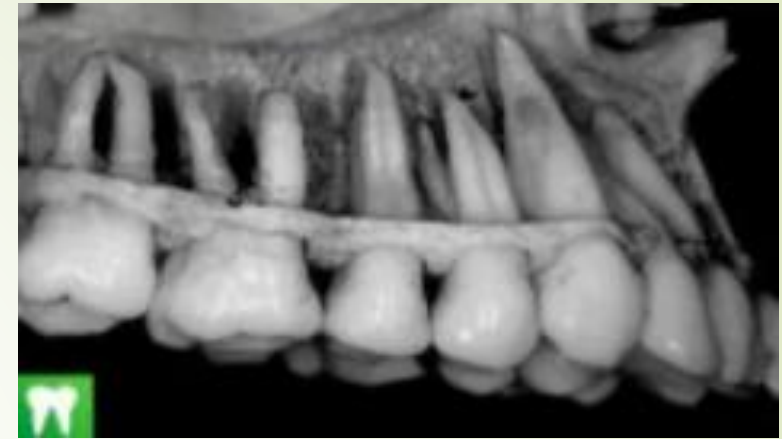
5. **Bone lining cells--Once osteoblasts have completed their function, they are either entrapped in the bone matrix and become osteocytes or remain on the surface as lining cells**

These cells flatten, when bone is not forming and extend along the bone surface

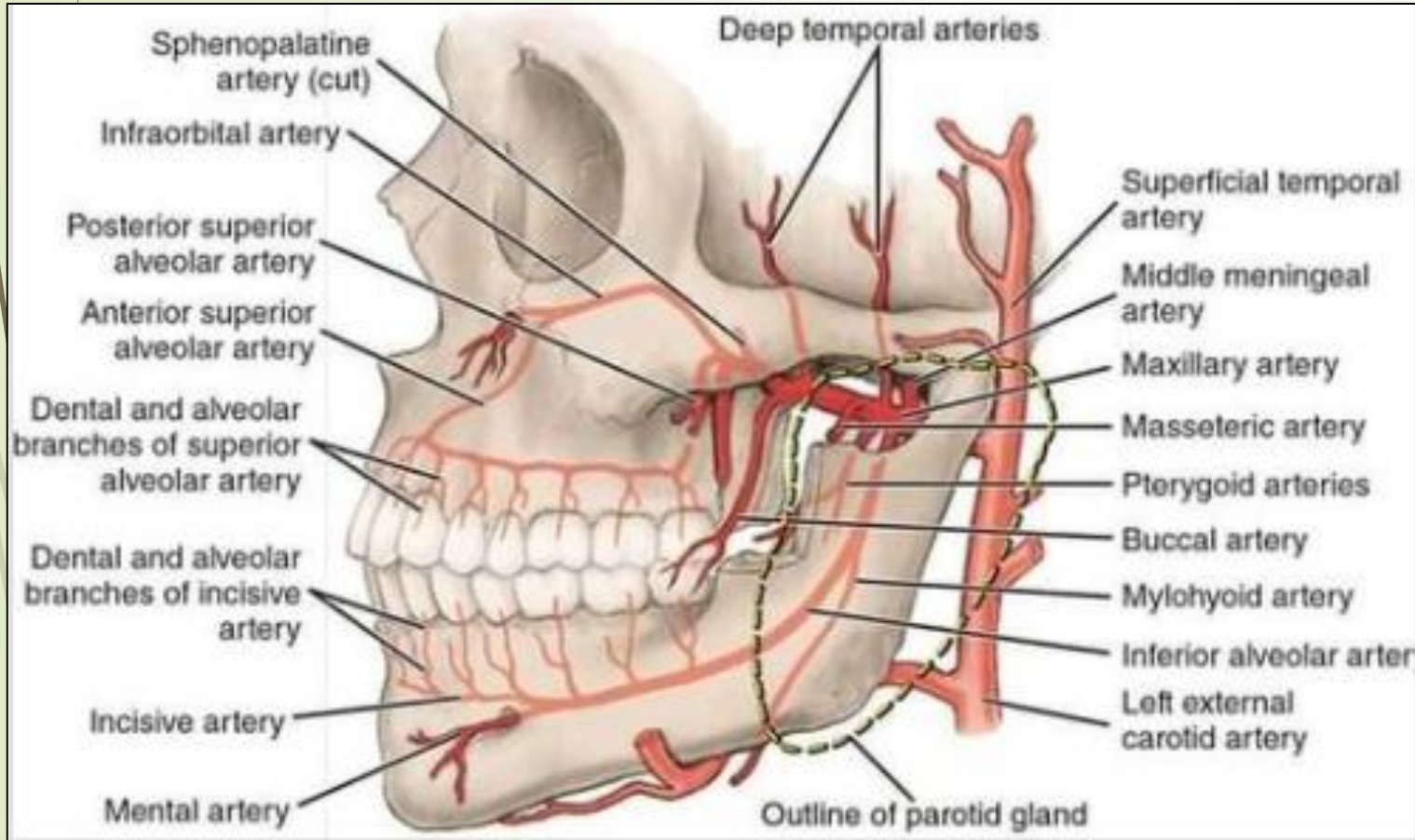


Functions of alveolar bone

- Houses the roots of teeth
- Anchors the roots of teeth to the alveoli, by the insertion of Sharpey's fibers into the alveolar bone proper
- Helps to absorb and distribute occlusal forces



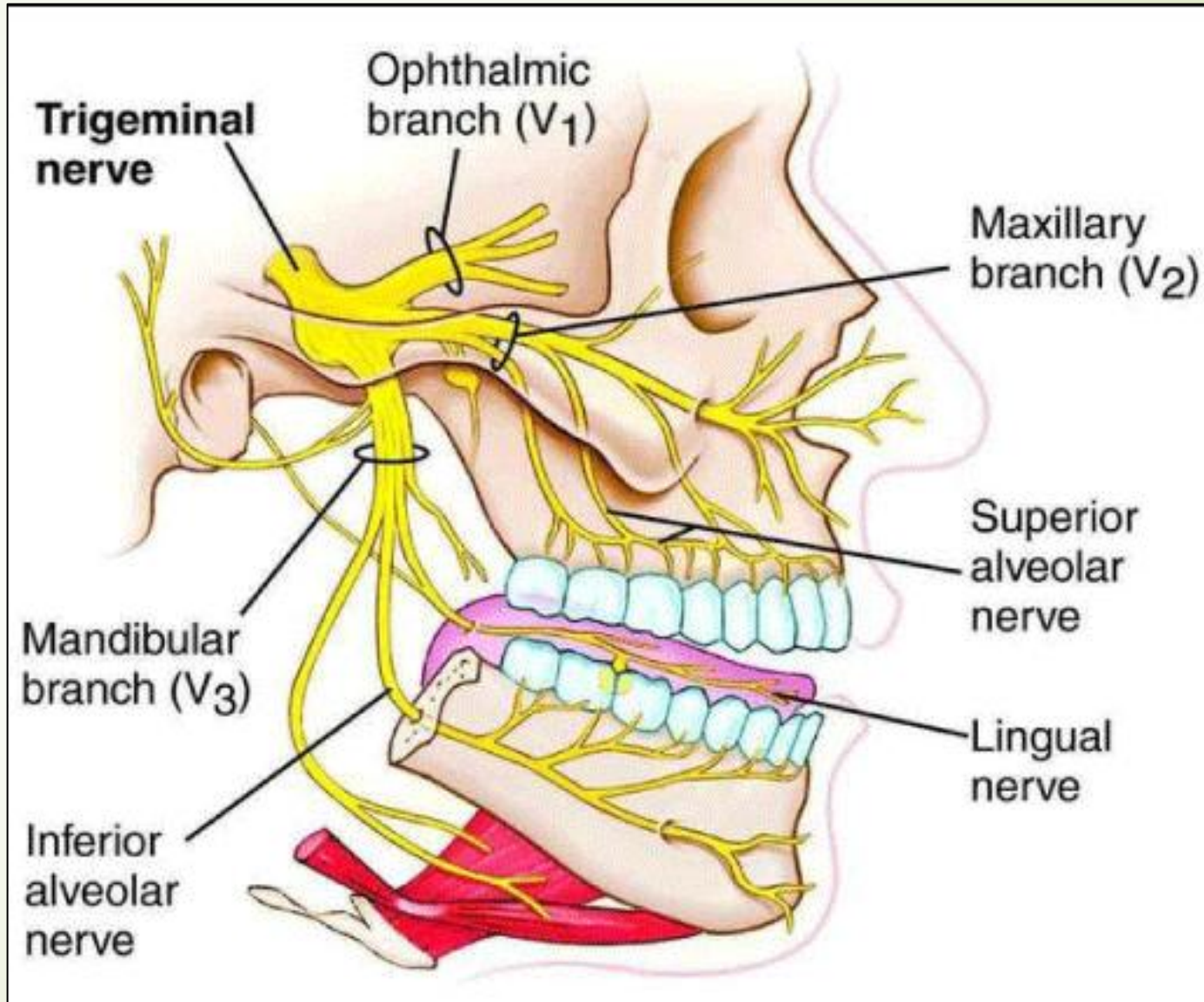
Blood supply of alveolar bone



- **Blood supply of maxilla—** branches of maxillary artery which is a branch of external carotid artery
- **Blood supply of mandible—** inferior alveolar artery which is a branch of maxillary artery

Nerve supply of alveolar bone

- Nerve supply of maxilla—superior alveolar (anterior, middle and posterior) and palatine branches of maxillary nerve
- Nerve supply of mandible—inferior alveolar nerve, a branch of mandibular nerve

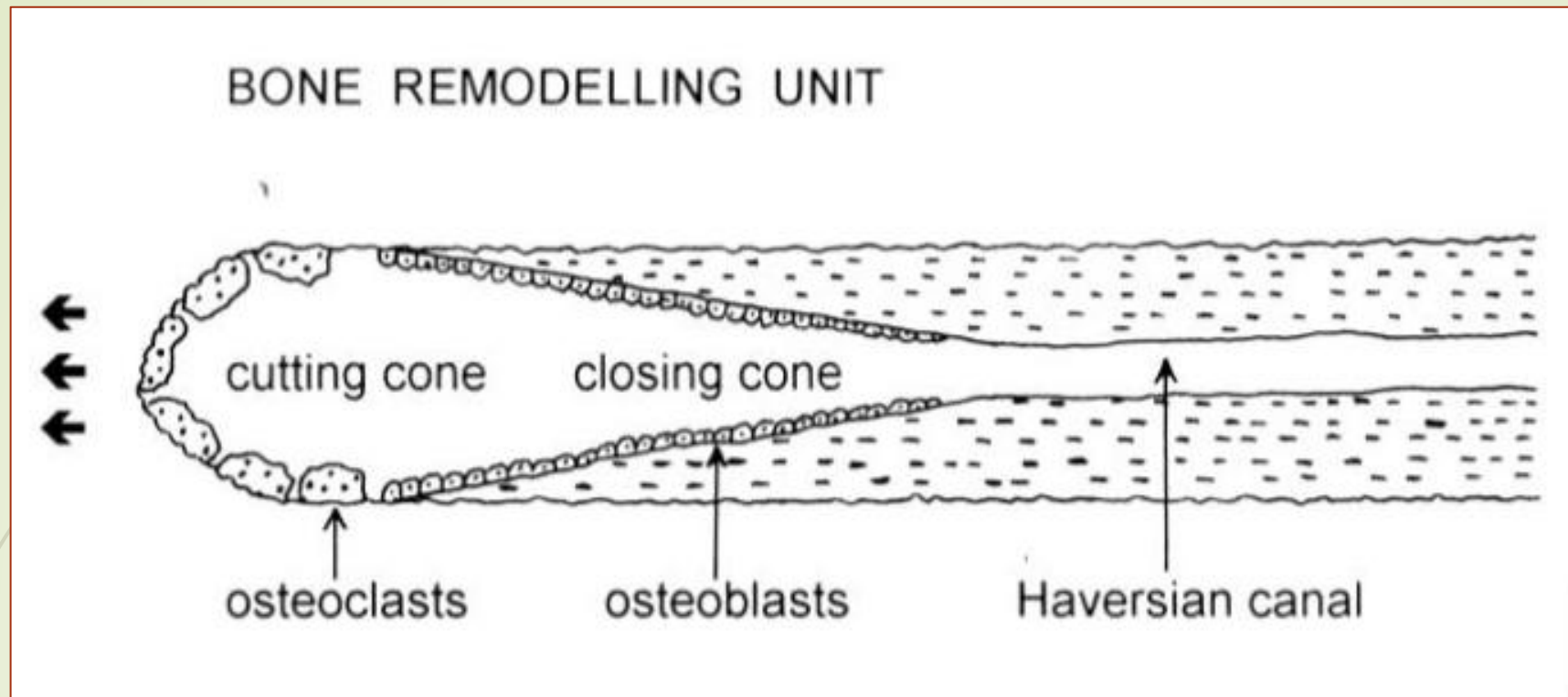


Remodeling of alveolar bone

- ***Bone modeling*** --process by which the overall size and shape of bones is established. It extends from embryonic bone development to the pre-adult period of human growth-- initial bone architecture
- **Bone is formed rapidly on the periosteal surface and destroyed along the endosteal surface**
- ***Bone turnover or remodeling*** --replacement of old bone by new bone
- **To maintain stability and integrity of bone, it constantly undergoes remodeling**

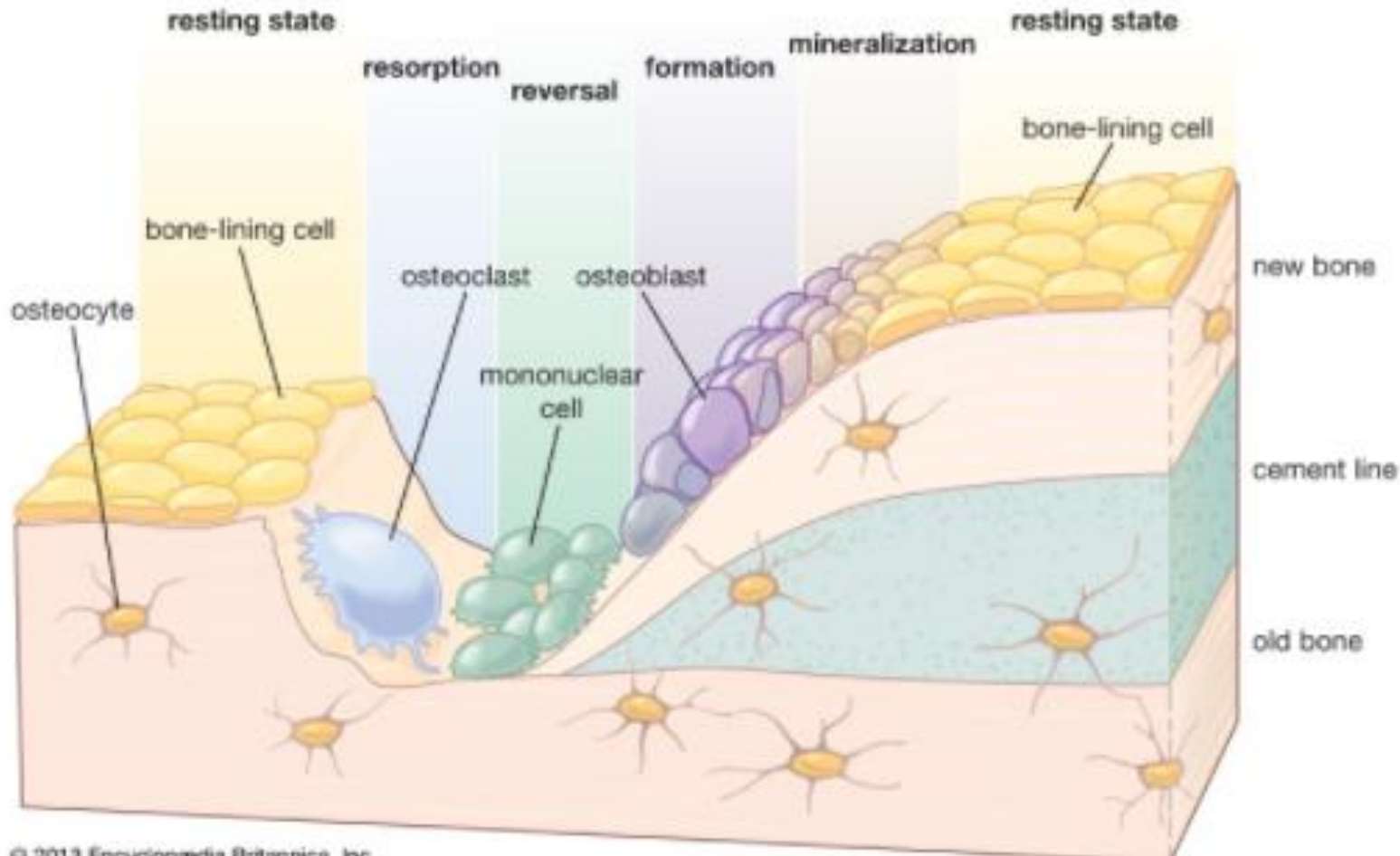
- **Brought about by osteoclasts and osteoblasts**
- **Osteoclasts resorb bone by tunneling through it-- called cutting cone characterized by a scalloped array of Howship's lacunae, each housing osteoclasts**
- **Behind the cutting cone, mononucleated migrate and differentiate into osteoblasts, and produce a coating termed the cement or reversal line**

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- **Cement line**-- a thin layer of glycoproteins comprising at least bone sialoprotein and osteopontin that acts as a cohesive, mineralized layer between the old bone and the new bone to be secreted.
 - On top of the cement line osteoblasts begin to lay down new bone matrix, mineralizing it from the outer to inner side--termed **the filling cone**.
 - As formation proceeds, some osteoblasts become osteocytes. Once complete, the Haversian canal contains a central blood vessel and a layer of inactive osteoblasts, the lining cells that communicate with osteocytes
 - Bone synthesis and bone breakdown go on simultaneously and the status of the bone is a net result of a balance between the two processes--called '**coupling**' of **bone resorption and formation**



Schematic diagram of a forming osteon or bone remodelling unit. Osteoclasts (bone resorbing cells) cut a longitudinal tunnel through existing cortical bone. The tunnel is subsequently infilled by osteoblasts (bone forming cells), which lay down new layers of bone arranged concentrically around a central space called a Haversian canal.

Bone remodeling

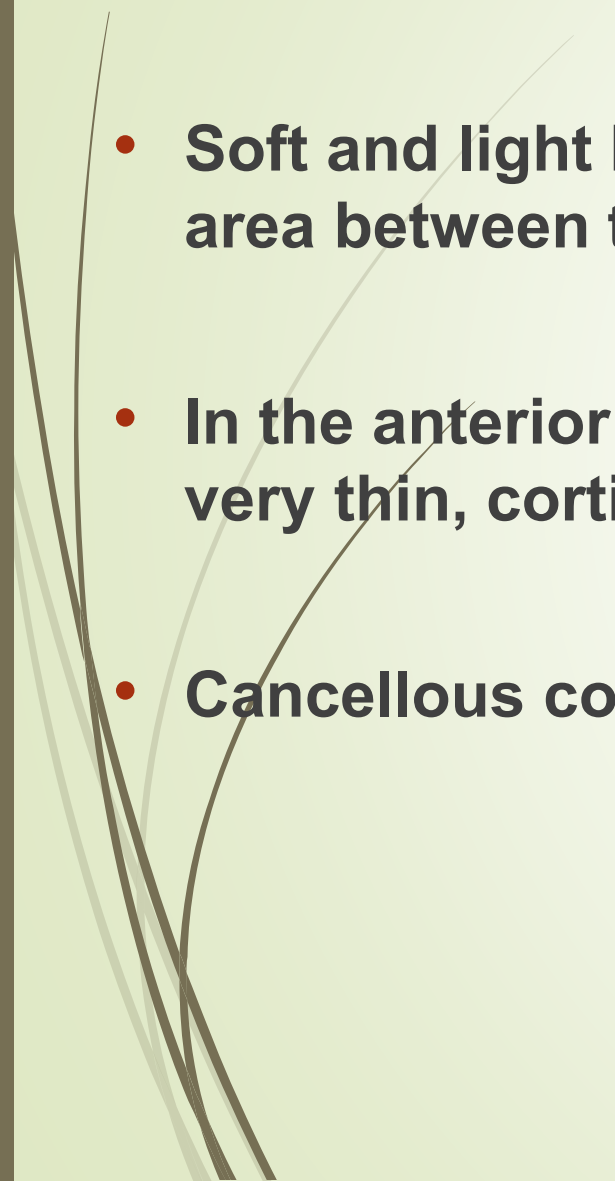


The remodeling process occurs through the cooperative activity of various cells that form basic multicellular unit or bone remodeling unit. The process begins with the activation of osteoclast formation, followed in order by

- (1) a resorption phase during which osteoclasts remove old bone and create the resorption lacuna,
- (2) reversal in which mononuclear cells (osteoblast precursors) deposit a cement line,
- (3) formation phase during which new bone is deposited, and finally
- (4) resting phase during which osteoblasts become quiescent and become the flattened bone lining cells.

Differences between cortical and cancellous bone

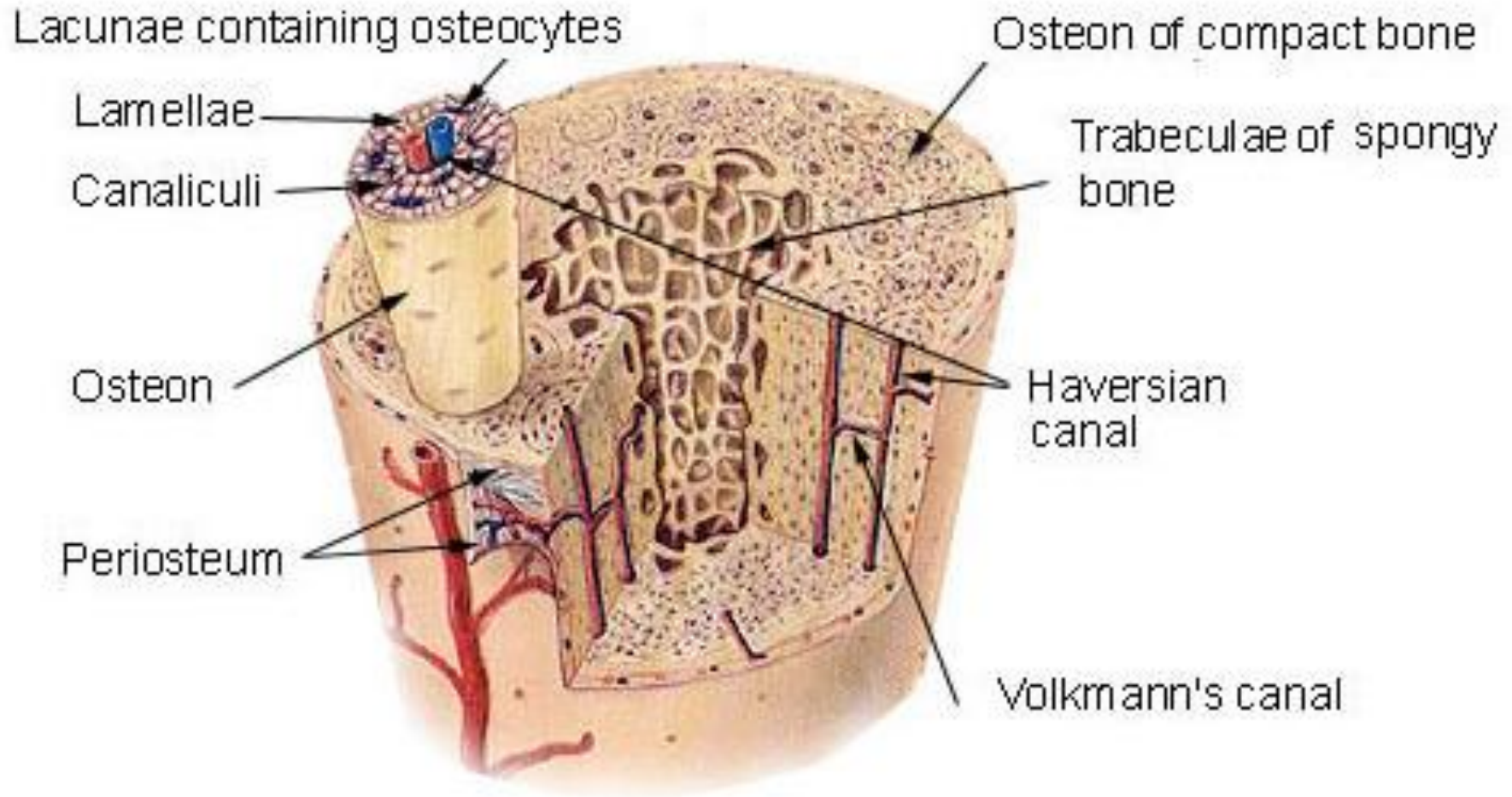
- **Compact bone consists of closely packed osteons /haversian systems (central canal surrounded by concentric rings (lamellae) of matrix)**
- **Between the rings of matrix, osteocytes are located in spaces called lacunae. Canaliculi radiate from the lacunae to haversian canal to provide passageways through the hard matrix**
- **In compact bone, the haversian systems are packed tightly together to form a solid mass**
- **The osteonic canals contain blood vessels that are parallel to the long axis of the bone**
- **Thinner in the maxilla, than in the mandible. Thickest in the premolar and molar region of the lower jaw, especially on the buccal side**

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- **Soft and light bone made up of loosely arranged trabeculae, fills the area between the cortical plates and the alveolar bone proper**
 - **In the anterior region of both jaws, the supporting bone usually is very thin, cortical plate is fused with the alveolar bone proper**
 - **Cancellous component in maxilla is more than in the mandible**

Cancellous bone

- **Spongy (cancellous) bone is lighter and less dense than compact bone. Spongy bone consists of plates (trabeculae) and bars of bone adjacent to small, irregular cavities that contain red bone marrow**
- **The canaliculi connect to the adjacent cavities, instead of a central haversian canal, to receive their blood supply**
- **Trabeculae are organized to provide maximum strength. The trabeculae of spongy bone follow the lines of stress and can realign if the direction of stress changes**

Compact Bone & Spongy (Cancellous Bone)



Differentiate histologically and radiographically between maxillary and mandibular alveolar bone

- Radiographically, the architecture **type I is seen most often in the mandible** whereas **type II in the maxilla**
- In the maxilla, the outer cortical plate is perforated by many small openings through which blood and lymph vessels pass

Age changes of alveolar bone with decrease in bone formation

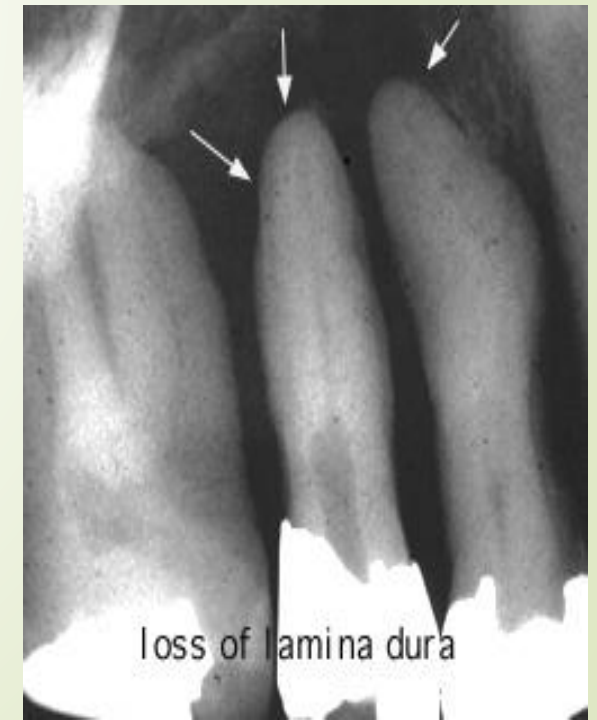
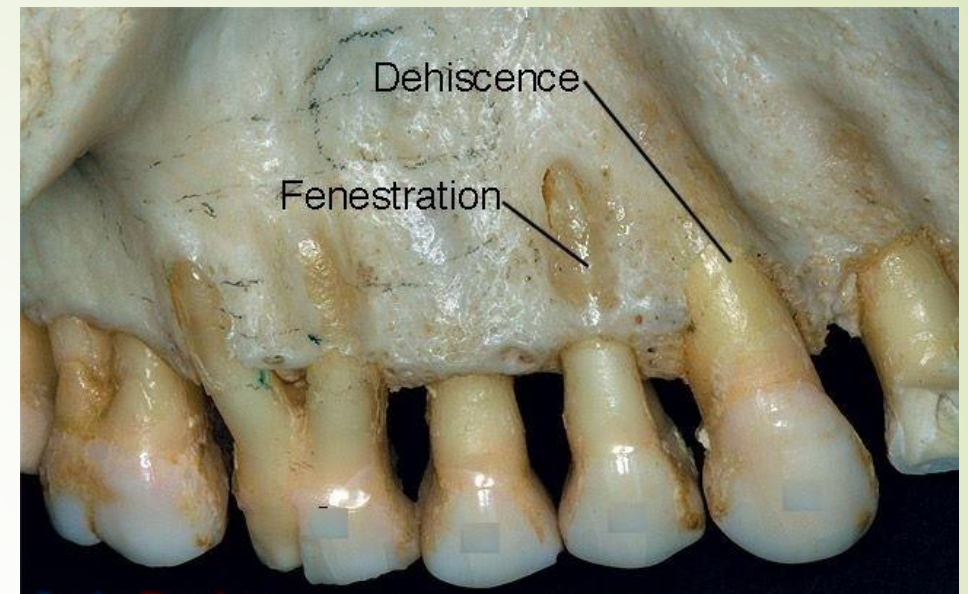
- Gradual decrease in bone formation with decrease in the bone mass—decreases osteoblast proliferating precursors or decreased synthesis and secretion of essential bone matrix proteins
- The width of cribriform plate and interdental alveolar septum decreases
- The marrow spaces show fatty infiltration

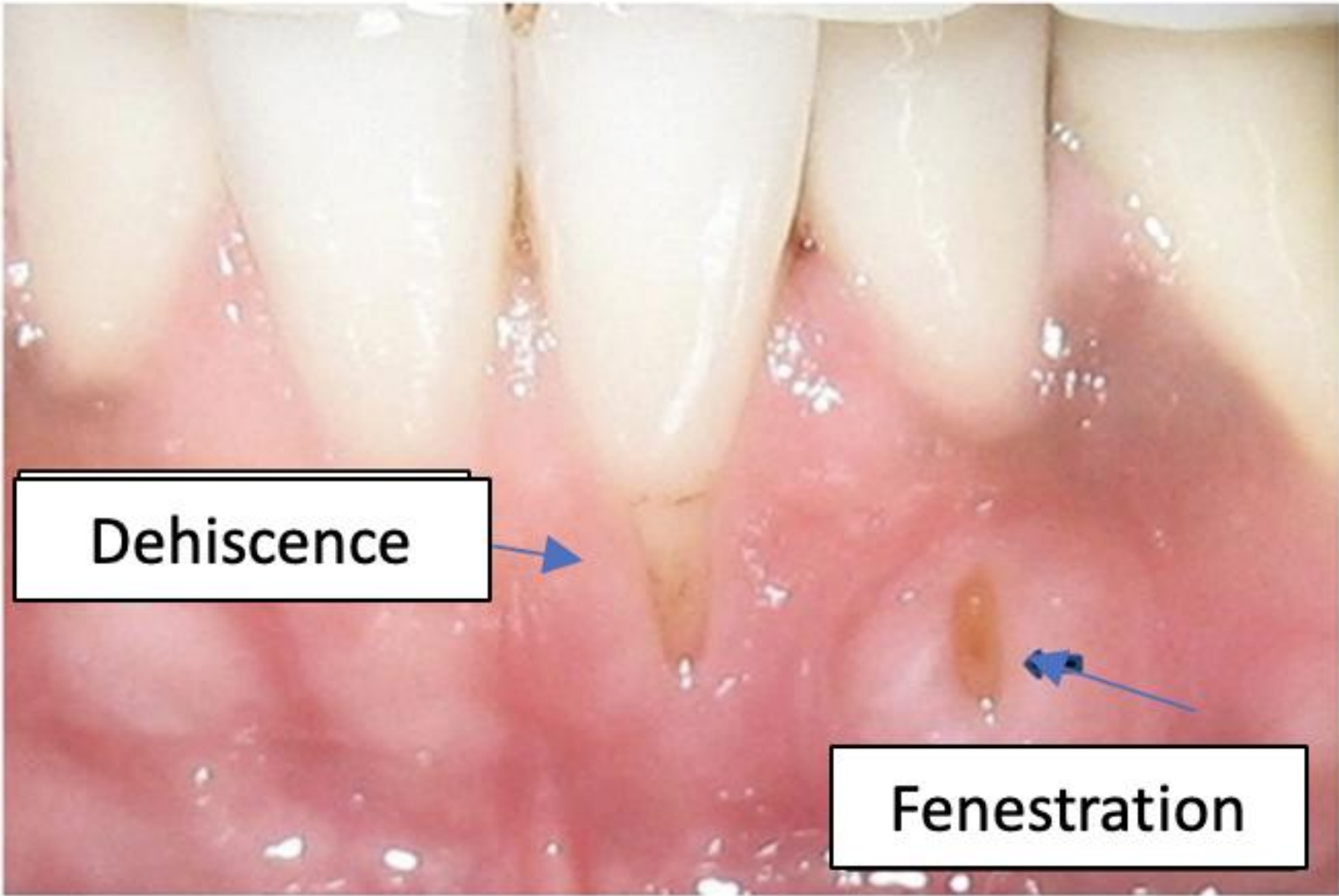
Clinical considerations

- **Bone is a highly plastic tissue--bone is resorbed on the side of pressure and apposed on the side of tension; thus the entire alveolus is allowed to shift with the tooth**
- **At sites of pressure, osteoclasts proliferate and cause resorption**
- **At sites of tension, osteoblasts are activated to produce osteoid and form new bone**
- **Alveolar bone loss may be due to –periodontal inflammation and tooth extraction**

Alveolar ridge defects and deformities can also be the result of congenital defects, trauma, periodontal disease or surgical ablation.

- **Fenestrations**(A localized bone defect that exposes the tooth root, but the alveolar crest (the top edge of the bone) remains intact.) **and dehiscences** (A bone defect where the loss of bone extends to the top edge of the alveolar crest.)
- **Loss of lamina dura-- infections, inflammation and resorption of bone socket**





Dehiscence

Fenestration

References

- 1. Orban's Oral Histology and Embryology-14th Edition. Chapter 9: Bone; Page no: 165-193**
- 2. Tencate's Oral Histology 8th edition. Chapter 6: Bone; Page no: 95-121**